Exchange rate and Production network

Macro Internal Seminar

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Motivation

Exchange rate → Imported inputs → Upstream firms → Downstream firms → Aggregate inflation

Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

This paper: both together

Does accounting for production networks change the optimal exchange-rate regime?

Literature

Network side (closed economy): Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

Open-economy side (no network): Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

Open economy with network (closest to this paper):

- **Gnocato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025)** — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice[1]
- **Qiu, Wang, Xu & Zanetti (2026, JME)** — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

Gap: network + open economy + FX regime choice, together.

[1] Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo $\rightarrow$ SOE: imported inputs Ω^F, UIP, debt-elastic premium

Ask: optimal FX regime? DC index survive with 2 cost-push channels?

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Not just Galí-Monacelli with sectors: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new

Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Households

Small open economy: home takes foreign price P_t^{F*}, rate i^*, and demand D_t^* as given.

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize lifetime utility, subject to a budget constraint with a domestic bond B_t and a foreign bond B_t^*:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition; CRS makes marginal cost common across firms:

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H}\, M_{ift}^{\Omega_i^F}$$

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} \left(S_t P_t^{F*}\right)^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) *and* S_t, the exchange rate (via the import cost $S_t P_t^{F*}$); with **Calvo pricing**, only a fraction δ_i resets each period.

Firms:Production Network

Stacking each firm's input shares Ω_{ij}^H, Ω_i^F across sectors gives the **input-output network** $\Omega^H \in \mathbb{R}^{N\times N}$, import-exposure vector $\boldsymbol{\omega}_F \in \mathbb{R}^N$ (import cost share of each sector's output), and export-exposure vector $\boldsymbol{\omega}_X \in \mathbb{R}^N$ (export share of each sector's own output).

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

applied to consumption shares $\boldsymbol{\beta}^H$ (Households) it gives each sector's **Domar weight**; applied to $\boldsymbol{\omega}_F$ or $\boldsymbol{\omega}_X$, its import or export centrality:

$$\boldsymbol{\lambda}_D^\top = (\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}, \qquad \mathcal{M} = (I - \Omega^H)^{-1}\boldsymbol{\omega}_F, \qquad \mathcal{M}^X = (I - \Omega^H)^{-1}\boldsymbol{\omega}_X$$

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H\Delta)^{-1}$, applied to labor share $\boldsymbol{\alpha}$ vs. import share $\boldsymbol{\omega}_F$:

$$\mathcal{B} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa}$$

Only relative shocks matter: $\mathcal{V}\mathbf{1} = \mathbf{0}$ — uniform TFP shocks create no cost-push; only sector-specific gaps $\boldsymbol{\chi}_t = (I - \Omega^H)^{-1}\log\mathbf{A}_t - \boldsymbol{p}_{t-1}$ do.

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t: exch. rate chg.	κ: GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Foreign Sector

Law of one price and UIP with a debt-elastic risk premium RP_t:

$$P_t^{FH} = S_t P_t^{F*}, \qquad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance; exports are **sector-specific** (Resource-heavy, matching Chile's data):

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \qquad EX_{it} = \kappa_{EX,i} D_t^* \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \;\rightarrow\; MC_{it} \;\rightarrow\; \pi_{it} \quad \text{(cost-push)} \qquad \hat{s}_t \;\rightarrow\; EX_{it} \;\rightarrow\; \tilde{y}_t \;\rightarrow\; \pi_{it} \quad \text{(demand)}$$

Monetary Policy Regimes

Three exchange-rate regimes are compared.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t\pi_t^C - r_t^{\mathrm{nat}}\right) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta\left(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H\right)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t, \qquad \hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H\sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t + \phi_s\hat{s}_t & \text{Managed} \end{cases}$$

where: r_t^{nat} natural real rate, $\hat{p}_t^H$ log home price idx., π_t^{DC} DC index (target); $\mathcal{B}, \Gamma, \mathcal{V}, \psi(\cdot), RP_t$ as before.

Welfare loss: $\mathcal{W} = \frac{1}{2}\left[(\gamma + \varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \kappa_i \mathrm{Var}(\pi_{it})\right]$, $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$

Calibration: Chile Data

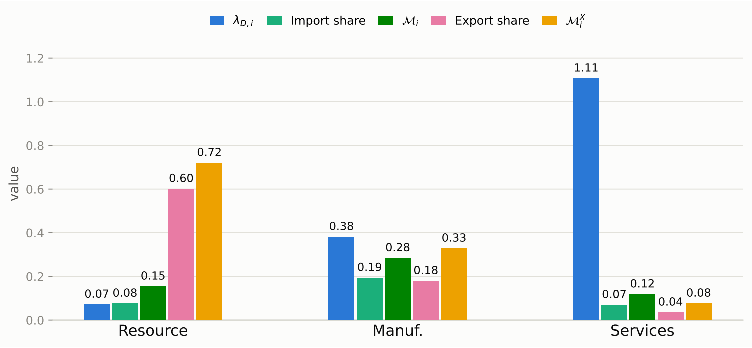
Parameter	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604	Data: BCCh IO table
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744	
Export share$_{1,2,3}$	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); Nakamura–Steinsson (2008)
ψ	0.020	Key mechanism: risk-premium/UIP elasticity & FX-regime rules (see ψ-sensitivity, appendix)
ϕ_π	1.50	
ϕ_y	0.50	
ϕ_s	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): β=0.99, γ=1.00, φ=2.00, η=1.50, ε=8.00, θ^*=2.00, $\beta^{F,tot}$=0.10.

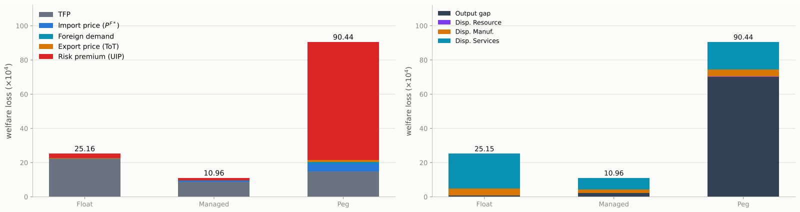
Network Properties: Import & Export Exposure



The production network amplifies both import and export exposure well beyond each sector's direct (raw) share — e.g. import exposure roughly doubles for every sector once indirect, network-propagated exposure is counted, not just each sector's own imports.

$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i$ (Domar weight), $\mathcal{M}_i = [(I - \Omega^H)^{-1}\boldsymbol{\omega}_F]_i$ and $\mathcal{M}_i^X = [(I - \Omega^H)^{-1}\boldsymbol{\omega}_X]_i$ (import/export centrality) — full notation: Appendix.

Welfare Ranking: Managed Float Dominates



Managed float dominates both corners: Float 25.15, Managed 10.96, Peg 90.45 ($\times 10^{-4}$, real Chile calibration) — Managed $<$ Float $\ll$ Peg, Peg no longer competitive

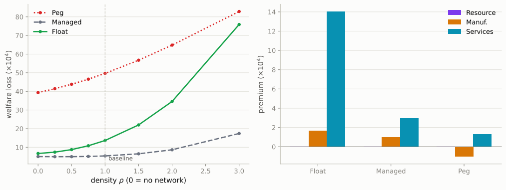
By shock (left): risk-premium/UIP dominates Peg (76% of its total). **By gap vs. dispersion (right):** Peg's loss is $\approx$77% output gap (70.01 of 90.45, zero monetary autonomy); Float's and Managed's are almost entirely price dispersion, concentrated in Services — full decomposition in appendix

Isolating the Network Channel

The causal experiment: Network OFF ($\Omega^H \equiv 0$) vs. Network ON (real Chile Ω^H) — same shocks, same regimes, only the domestic input-output structure changes.

Total welfare loss ($\times 10^4$)	**Float**	**Managed**	**Peg**
Network OFF	8.67	5.82	33.66
Network ON (baseline)	25.15	10.96	90.45
% increase from the network	**+190%**	**+88%**	**+169%**

The network substantially increases welfare losses under Float and Peg but barely affects Managed — and Managed stays the best regime either way.



Continuous density dial ρ (stylized network) confirms it: losses rise **monotonically** with density, and the Peg/Float gap **widens** as the network gets denser.

Mechanism: How the Network Amplifies Shocks

Output-gap variance, **Network ON** vs. **Network OFF** (real Chile calibration) — full IRFs: Appendix:

Regime	Import-price shock Var($\tilde{y}_t$) ratio, ON/OFF	Risk-premium/UIP shock Var($\tilde{y}_t$) ratio, ON/OFF
Float	5.4×	8.5×
Managed	6.0×	1.1× (*essentially none*)
Peg	11.1×	4.8×

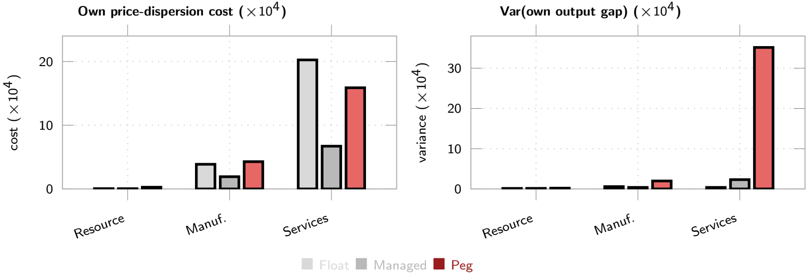
Amplification is regime-and-shock-specific, not a generic network multiplier. Import-price shocks hit **Peg** hardest (no FX buffer to absorb pass-through); risk-premium shocks hit **Float** hardest ($\hat{s}_t$ free to move transmits the shock straight through the network) — **Managed** blocks amplification of the risk-premium shock almost entirely, but not the import-price shock

DC inflation variance is unchanged or slightly lower with the network on, for *both* shocks in every regime — this isn't new cost-push, it's a redirection of the existing shock into output

Takeaway: which shock dominates a regime's loss (Welfare Ranking, above) depends on which margin that regime leaves free to adjust

Network Exposure & Sector Welfare Preferences

Different sectors prefer different exchange-rate regimes, in both price and quantity. A representative-sector model cannot generate this: it only exists because sectors differ in Calvo stickiness $\hat{\delta}_i$ and network centrality $\lambda_{D,i}$, so the same shock hits their welfare weight $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ and their output-gap exposure differently.



Left (price dispersion): Managed dominates every sector. Peg costliest for Resource & Manuf.; for Services, **Float is costliest** (20.25 > Peg's 15.88) — near-unit-root S_t under Float feeds the stickiest sector's marginal cost for a long time. Aggregate ranking is set by Services, which carries most of the economy's welfare weight

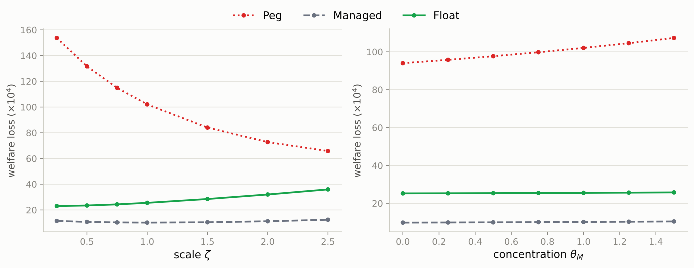
Right (output-gap variance, descriptive — the loss function only weights the aggregate gap): opposite pattern — **Peg is the outlier**, concentrated almost entirely in Services (35.2); Peg's cost is almost entirely a *quantity* story, Float's Services cost almost entirely a *price*-dispersion one

Openness & Import Concentration

Reparametrize $\boldsymbol{\omega}_F$ only, on the REAL Chile calibration ($\alpha_i = 1 - \Omega_i^H - \omega_{F,i}$ absorbs it, holding Ω^H fixed):

$$\omega_{F,i}(\zeta) = \zeta\, \omega_{F,i}^{\text{base}} \qquad \omega_{F,i}(\theta_M) = \bar{\omega}_F + \theta_M \left(\omega_{F,i}^{\text{base}} - \bar{\omega}_F\right)$$

$\zeta = \theta_M = 1$: baseline. ζ: uniform rescaling. θ_M: mean-preserving redistribution (θ_M=0: uniform exposure).



Managed < Float < Peg robust throughout (aggregate-export specification); gap moves in **opposite directions**:

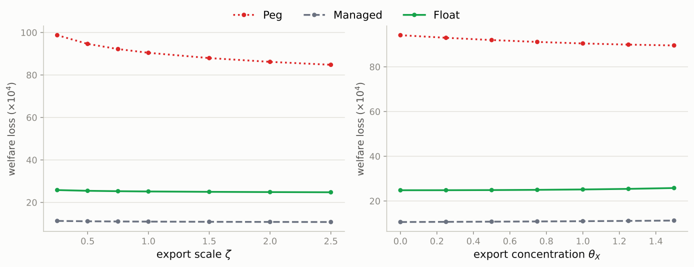
- ζ (openness): Peg ↓ (153.7 → 65.8), Float ↑ (23.1 → 36.0) — **converge**
- θ_M (concentration): Peg ↑ faster (94.0 → 107.3) than Float (25.2 → 25.8) — **gap diverges**, via Peg rising, not Float falling
- **Mechanism:** $\alpha_i + \Omega_i^H + \omega_{F,i} = 1$ links labor- and import-cost channels; Peg's risk-premium/output-gap channel is far more sensitive to reallocation toward Services than Float's price-dispersion channel

Export Openness & Export Concentration

Same construction, applied to the sector-export model's $\kappa_{EX,i}$ (real Chile calibration; $\zeta = \theta_X = 1$ recovers 25.15/10.96/90.45 exactly):

$$\kappa_{EX,i}(\zeta) = \zeta\, \kappa_{EX,i}^{\text{base}} \qquad \kappa_{EX,i}(\theta_X) = \bar{\kappa}_{EX} + \theta_X \left(\kappa_{EX,i}^{\text{base}} - \bar{\kappa}_{EX}\right)$$

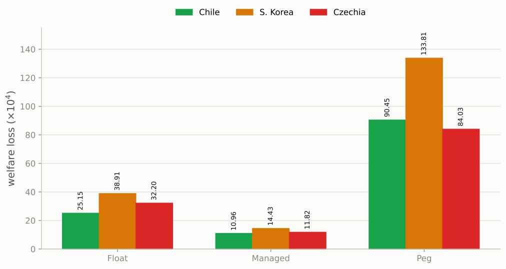
$\zeta = \theta_X = 1$: baseline. ζ: uniform export-openness scaling. θ_X: mean-preserving redistribution (θ_X=0: uniform export exposure).



Managed < Float < Peg robust throughout. Unlike import ζ (opposite directions — $\omega_{F,i}$ trades off against α_i), export ζ moves **all three regimes together**:

- ζ (openness): **all regimes improve** as trade opens — Peg most (98.7 → 84.8), since $\kappa_{EX,i}$ only enters export demand, not the cost side
- θ_X (concentration): **opposite directions, as with import concentration** — Peg ↓ (94.1 → 89.6), Float/Managed ↑ slightly, as exports concentrate toward Resource (the flexible-price sector)
- **Mechanism:** routing exports toward Resource's low κ_1 helps Peg (flexible-priced sector absorbs shocks better); raises Float/Managed's dispersion cost slightly instead

Robustness: South Korea & Czechia



The ranking is robust: Managed $<$ Float $\ll$ Peg in **every** calibration. UIP dominates Peg's loss throughout (76% Chile; 72%/65% Korea/Czechia). Also robust to a 2nd-order solution and to the import/export openness & concentration sweeps above.

Why Korea is uniformly higher (Float and Peg): denser network (Ω^H row sums 0.50/0.50/0.39 vs. Chile's 0.42/0.45/0.33) *and* **higher import exposure**, esp. Manufacturing (Ω_2^F=0.24 vs. 0.195, electronics/machinery supply chains) $\Rightarrow$ import centrality $\mathcal{M}_i$ higher in every sector (Manuf. 0.41 vs. 0.29) $\Rightarrow$ bigger Γ (Float's price-dispersion channel) *and* more network amplification of the output gap (Peg's channel) — same two levers as "Isolating the Network Channel," just further along the dial.

Note: Korea/Czechia use the aggregate-export specification, so comparisons across countries are illustrative of network/calibration robustness rather than a like-for-like export comparison.

Conclusion

Yes, networks change how the exchange rate operates: FX pass-through and TFP cost-push travel through the *same* rigidity-adjusted network operator — once S_t is a second cost-push channel, the closed-economy divine-coincidence index generically fails

Networks matter for severity and incidence, not the ranking: switching the domestic network on substantially raises welfare losses in every regime and reallocates which sector bears them — but **Managed** $<$ **Float** $<$ **Peg either way**

Peg is dominated by a different mechanism than the textbook story: not terms-of-trade, but risk-premium/UIP — zero monetary autonomy forces the shock straight into the output gap

Managed float wins because it is the only regime that dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

Robust across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution